

MD RAKIB HASAN

Food Engineering | Food Innovation | Sustainable Food Systems | Food Packaging

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PERSONAL PROFILE

Food Engineering and Food Innovation graduate with academic research experience and over three years of industrial R&D experience in beverage and dairy product development. Experienced in food formulation, experimental trials, shelf-life assessment, sensory evaluation, functional foods, food packaging research and laboratory analysis. Academic research experience includes biodegradable packaging films, smart packaging and consumer research, alongside academic learning in alternative protein systems and sustainable food innovation. Interested in doctoral research at the intersection of sustainable food systems, alternative proteins, innovative functional food products and advanced food packaging technologies.

RESEARCH INTERESTS

Alternative proteins and sustainable protein systems • Plant-based and next-generation food product development • Sustainable and biodegradable food packaging • Smart and active food packaging • Sustainable food systems and food innovation • Functional food • Food quality, functionality and shelf-life • Consumer acceptance of emerging food technologies

EDUCATION

University of Greenwich, United Kingdom

MSc Food Innovation | 2025–2026

Dissertation: A Quantitative Study of Consumer Perceptions towards Smart Packaging in Processed Food Items in Bangladesh

- Conducted quantitative research investigating consumer awareness, attitudes and behavioural intentions towards smart packaging technologies.
- Designed and conducted a structured survey using Qualtrics and developed the research methodology for quantitative data collection.
- Gained experience in research design, literature review, data collection and statistical analysis using IBM SPSS.
- Developed academic knowledge of food innovation, emerging food technologies and sustainable food systems.

Relevant Module: Alternative Proteins

- Developed knowledge of emerging alternative protein sources and their applications in food product development.
- Studied technological, nutritional and sustainability considerations associated with alternative protein systems.
- Explored formulation and processing challenges in alternative protein-based foods, including functionality, texture and consumer acceptance.

MSc Food & Process Engineering | 2021–2022

Hajee Mohammad Danesh Science & Technology University (HSTU), Dinajpur, Bangladesh

BSc Food & Process Engineering | 2016–2021

Hajee Mohammad Danesh Science & Technology University (HSTU), Dinajpur, Bangladesh

RESEARCH EXPERIENCE

Research Assistant | Institute of Research and Training, Department of Food Engineering and Technology, HSTU

Project: Preparation of Synthetic and Extracted Gelatin-CMC Based Packaging Films and Experimentation for Packing Baked Products

- Participated in the development and laboratory evaluation of gelatin-CMC-based biodegradable packaging films for food applications.
- Assisted with preparation, experimentation and evaluation of packaging-film samples.
- Investigated the potential application of biodegradable films for packaging baked food products.
- Conducted laboratory experiments and maintained experimental records and research documentation.
- Project funded by the University Grants Commission of Bangladesh.

RESEARCH PROJECT

Effect of Microwave Heating on Quality Attributes of Sugarcane Juice During Refrigerated Storage

- Investigated the effect of microwave heating on selected quality attributes of sugarcane juice during refrigerated storage.

- Conducted experimental work to evaluate changes in product quality during storage.
- Analysed and documented experimental observations and contributed to interpretation of research findings.
- Contributed to preparation of the research manuscript.

INDUSTRIAL EXPERIENCE

Akij Food & Beverage Ltd. (AFBL), Bangladesh | Research & Development Executive | October 2022 – January 2025

- Developed and optimized formulations for beverage and dairy products through laboratory trials and product evaluation.
- Conducted R&D trials for products including Clemon Mojito, Akij Mango Fruit Drink and Flavoured Electrolyte Drink.
- Standardized formulation trial records, technical reports and documentation for new product development.
- Developed manufacturing instructions for production lines and operating procedures for laboratory and process-line equipment.
- Conducted sensory evaluation and shelf-life studies to assess product quality and stability.
- Collaborated with production and quality teams during product development, scale-up and technical problem-solving.
- Supported formulation optimization and improvement of functional food products.

TECHNICAL & RESEARCH SKILLS

Research: Experimental design, literature review, quantitative research, survey design, data collection, laboratory experimentation, research documentation, data interpretation, shelf-life studies

Food Technology: Functional food formulation, product development, food processing, sensory evaluation, food packaging, food quality assessment, alternative protein systems

Laboratory: Kjeldahl system, spectrophotometer, viscometer, rheometer, antioxidant analysis

Research & Software: Qualtrics, IBM SPSS, Microsoft Excel, Microsoft Office

Food Safety & Quality: HACCP, GMP, GLP, ISO 22000, FSSC, food quality management

PUBLICATIONS & RESEARCH OUTPUT

Publication: EFFECT OF TEMPERATURE ON DRYING KINETICS, PHYSICO-CHEMICAL PROPERTIES AND ANTIOXIDANT ACTIVITY OF CABINET DRIED WHOLE JUJUBES | **Article DOI:**

<https://doi.org/10.53430/ijmru.2025.9.2.0021>

Manuscript under review: Effect of Microwave Heating on Quality Attributes of Sugarcane Juice During Refrigerated Storage. Submitted to the Journal of Food and Nutrition Research.

Research Project: Preparation of Synthetic and Extracted Gelatin-CMC Based Packaging Films and Experimentation for Packing Baked Products. University Grants Commission-funded research project, HSTU.

CERTIFICATIONS & TRAINING

- RSPH Level 3 HACCP / Food Safety
- ISO 22000:2018 Food Safety Management System Training
- Productivity Improvement Tools and Techniques, National Productivity Organization (NPO), Ministry of Industries, Bangladesh, 2020

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Founding Member – Skill Development Club of HSTU
- Students' Welfare Secretary – Food Engineers' Association, HSTU.

ACADEMIC REFERENCES

Joysree Roy
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